2.22 Differentiation: The Chain Rule

Worked Solutions 1–25

1 a)

$\frac{dy}{dx} = 3(4x)(2x^2+1)^2 = 12x(2x^2+1)^2$

1 b)

$\frac{dy}{dx} = 4(6x^2-5)(2x^3-5x)^3$

1 c)

$\frac{dy}{dx} = \frac{1}{2}(12x^3)(3x^4-1)^{-1/2} = 6x^3(3x^4-1)^{-1/2}$

1 d)

$\frac{dy}{dx} = -\frac{3}{2}(-2x)(1-x^2)^{-5/2} = 3x(1-x^2)^{-5/2}$

1 e)

$y = (5+x^{1/2})^{-1}$ so $\frac{dy}{dx} = -(\frac{1}{2}x^{-1/2})(5+x^{1/2})^{-2}$

1 f)

$y = (x^{-1}-x)^7$ so $\frac{dy}{dx} = 7(-x^{-2}-1)(x^{-1}-x)^6$

2)

$y = (5x^2-2x+1)^{1/2}$ so $\frac{dy}{dx} = \frac{1}{2}(10x-2)(5x^2-2x+1)^{-1/2}$

$$\left.\frac{dy}{dx}\right|_{x=2} = \frac{9}{\sqrt{17}}$$

3)

$y = (1+7x^2)^{1/2}$ so $\frac{dy}{dx} = \frac{1}{2}(14x)(1+7x^2)^{-1/2}$

$$\left.\frac{dy}{dx}\right|_{x=3} = \frac{21}{8}$$

At $x=3$, $y=8$.

$$y-8 = \tfrac{21}{8}(x-3) \Rightarrow 8y-64 = 21x-63 \Rightarrow 8y-21x = 1$$

4)

$y = (4-3x^2)^{-1/2}$ so $\frac{dy}{dx} = -\frac{1}{2}(6x)(4-3x^2)^{-3/2}$

$$\left.\frac{dy}{dx}\right|_{x=-1} = 3$$

At $x=-1$, $y=1$.

$$y-1 = -\tfrac{1}{3}(x+1) \Rightarrow 3y-x = 4$$

5 a)

$$\frac{dy}{dx} = 4(6x)(3x^2-1)^3 = 24x(3x^2-1)^3$$

Stat pt when $\frac{dy}{dx} = 0$.

$$16x(3x^2-1)^3 = 0 \Rightarrow x = 0 \text{ or } x = \pm\tfrac{1}{\sqrt{3}}$$

$(-\frac{1}{\sqrt{3}}, 0)$, $(0,1)$ and $(\frac{1}{\sqrt{3}}, 0)$

5 b)

$y = (1+x^2)^{-1/2}$ so $\frac{dy}{dx} = -\frac{1}{2}(2x)(1+x^2)^{-3/2}$ Stat pt when $\frac{dy}{dx} = 0$.

$$-\tfrac{1}{2}(2x)(1+x^2)^{-3/2} = 0 \Rightarrow x = 0$$

Stat pt is $(0,1)$.

6 a)

$\frac{dx}{dy} = 6y - 1$ so $\frac{dy}{dx} = \frac{1}{6y-1}$

6 b)

$\frac{dx}{dy} = 8y^3$ so $\frac{dy}{dx} = \frac{1}{8y^3}$

6 c)

$\frac{dx}{dy} = \frac{1}{4}(6y)(3y^2-1)^{-3/4}$ so $\frac{dy}{dx} = \frac{2}{3y}(3y^2-1)^{3/4}$

7 a)

$$\frac{dx}{dy} = -5(3-y)^5 \text{ so } \left.\frac{dx}{dy}\right|_{y=2} = -5 \quad \therefore \left.\frac{dy}{dx}\right|_{y=2} = -\tfrac{1}{5}$$

7 b)

$$\frac{dx}{dy} = \tfrac{1}{2}(2y-2)(y^2-2y+1)^{-1/2}$$

$$\left.\frac{dx}{dy}\right|_{y=4} = \tfrac{1}{2} \times 6 \times \tfrac{1}{3} = 1 \quad \therefore \left.\frac{dy}{dx}\right|_{y=4} = 1$$

7 c)

When $x = \frac{1}{2}$, $\sqrt{y^4+3} = 2 \Rightarrow y = 1$.

$$x = (y^4+3)^{-1/2}, \quad \frac{dx}{dy} = -\tfrac{1}{2}(4y^3)(y^4+3)^{-3/2}$$

$$\left.\frac{dx}{dy}\right|_{y=1} = -\tfrac{1}{2}(4)(1+3)^{-3/2} = -\tfrac{1}{4} \quad \therefore \left.\frac{dy}{dx}\right|_{y=1} = -4$$

8)

$$\frac{dy}{dx} = \tfrac{1}{2}(2x)(x^2+5)^{1/2} \text{ so } \left.\frac{dy}{dx}\right|_{x=2} = 6$$

At $x = 2$, $y = 9$.

$$y - 9 = 6(x-2) \Rightarrow y = 6x - 3$$

9 a)

$$y = x^{1/6} + 2$$

$$\frac{dx}{dy} = 6(y-2)^5 = 6(x^{1/6}+2-2)^5 = 6x^{5/6}$$

9 b)

So $\frac{dy}{dx} = \frac{1}{6x^{5/6}} = \frac{1}{6}x^{-5/6}$

10)

$$\frac{dy}{dx} = (1+3x-2x^2)^{-3}$$

$$\frac{d^2y}{dx^2} = -3(3-4x)(1+3x-2x^2)^{-4} = \frac{12x-9}{(1+3x-2x^2)^4}$$

Pt of inflection when $\frac{d^2y}{dx^2} = 0 \Rightarrow 12x - 9 = 0$ therefore at $x = \frac{3}{4}$.

11 a)

$y = (\sin x)^2$ so $\frac{dy}{dx} = 2\cos x(\sin x)^1 = 2\cos x \sin x$

11 b)

$y = (\cos x)^2$ so $\frac{dy}{dx} = -2\sin x(\cos x)^1 = -2\sin x \cos x$

11 c)

$\frac{dy}{dx} = 6\cos(3x-7)$

11 d)

$\frac{dy}{dx} = \frac{\pi}{5}\sin(\pi x - \frac{\pi}{2})$ (or $-\frac{\pi}{5}\cos(\pi x)$)

11 e)

$\frac{dy}{dx} = -\frac{1}{2}x^{-1/2}\sin(\sqrt{x})$

11 f)

$\frac{dy}{dx} = -6x^2\cos(x^3)$

11 g)

$\frac{dy}{dx} = \cos x \times -\sin(\sin x) = -\cos x \sin(\sin x)$

11 h)

$\frac{dy}{dx} = \cos x \times \cos(\sin x) = \cos x \cos(\sin x)$

11 i)

$y = (\sin x)^{1/2}$ so $\frac{dy}{dx} = \frac{1}{2}\cos x(\sin x)^{-1/2}$

11 j)

$y = (\cos x)^{1/3}$ so $\frac{dy}{dx} = -\frac{1}{3}\sin x(\cos x)^{-2/3}$

11 k)

$y = (\cos(5x))^3$ so $\frac{dy}{dx} = 5 \times \sin(5x) \times 3(\cos(5x))^2 = 15\sin(5x)\cos^2(5x)$

12)

$$\begin{aligned}\frac{d}{dx}[(\sin x)^{-1}] &= -\cos x(\sin x)^{-2} = -\frac{\cos x}{\sin^2 x} \\ &= -\frac{1}{\sin x} \times \frac{\cos x}{\sin x} = -\operatorname{cosec} x \cot x \text{ as req.}\end{aligned}$$

13 a)

Max and min are $2.85 + 0.75 = 3.6$ and $2.85 - 0.75 = 2.1$ respectively.

13 b)

$$\frac{dx}{dt} = 0.75 \times \frac{17\pi}{120}\cos\left(\frac{17\pi}{120}t + \frac{3\pi}{2}\right)$$

Stat pt when $\frac{dx}{dt} = 0$.

$$0.75 \times \frac{17\pi}{120}\cos\left(\frac{17\pi}{120}t + \frac{3\pi}{2}\right) = 0 \Rightarrow \cos\left(\frac{17\pi}{120}t + \frac{3\pi}{2}\right) = 0$$

$$\frac{17\pi}{120}t + \frac{3\pi}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots$$

$$t = 0, \frac{120}{17}, \frac{240}{17}, \frac{360}{17}, \ldots$$

So at $(0, 2.1)$, $(\frac{120}{17}, 3.6)$, $(\frac{240}{17}, 2.1)$, $(\frac{360}{17}, 3.6)$. Maximums at $\frac{120}{17}$ and $\frac{360}{17}$ therefore 07:04 and 21:11.

14)

$y = a(\sin x)^2 + b(\cos x)^2$ so $\frac{dy}{dx} = 2a\cos x \sin x - 2b\sin x \cos x$

$$\left.\frac{dy}{dx}\right|_{x=\pi/4} = a - b$$

But since $a \neq b$ the gradient at $x = \frac{\pi}{4}$ is never zero.

15)

$$\frac{dt}{dx} = 2.7 \times \frac{2\pi}{365} \cos\left(\frac{2\pi}{365}(x-78)\right)$$

$$\left.\frac{dt}{dx}\right|_{t=17} = 0.0231236 \text{ and } \left.\frac{dt}{dx}\right|_{t=85} = 0.0461413$$

The rate at which the number of daylight hours is increasing is higher on day85 compared to on day17.

16)

$$\mathrm{f}'(x) = \tfrac{\pi}{2} e^x \cos(\tfrac{\pi}{2} e^x)$$

Stat pt when $\mathrm{f}'(x) = 0$.

$$\tfrac{\pi}{2} e^x \cos(\tfrac{\pi}{2} e^x) = 0 \text{ but } e^x \neq 0 \Rightarrow \cos(\tfrac{\pi}{2} e^x) = 0$$

$$\tfrac{\pi}{2} e^x = \tfrac{\pi}{2}, \tfrac{3\pi}{2}, \tfrac{5\pi}{2}, \tfrac{7\pi}{2}, \ldots$$

So $e^x = 1, 3, 5, 7, \ldots \Rightarrow x = \ln 1, \ln 3, \ln 5, \ln 7, \ldots$ Therefore $x = \ln k$ where k is positive odd integer.

17)

$$\mathrm{f}(x) = \tfrac{1}{2}x + \cos(2x^{-1})$$

So $\mathrm{f}'(x) = \frac{1}{2} + 2x^{-2}\sin(2x^{-1}) = \frac{1}{2} + \frac{2}{x^2}\sin(\frac{2}{x})$. Increasing function when $\mathrm{f}'(x) > 0$.

$$\tfrac{1}{2} > \frac{2}{x^2}\sin(\tfrac{2}{x}) \Rightarrow x^2 > 4\sin(\tfrac{2}{x})$$

Largest $\sin(\frac{2}{x})$ can be is1.

$$x^2 > 4 \Rightarrow x^2 - 4 > 0 \Rightarrow (x-2)(x+2) > 0$$

Which is true for $x < -2$ and $x > 2$. Therefore always increasing for $x > 2$.

18 a)

$\frac{dy}{dx} = 10e^{2x}(e^{2x}-1)^4$

18 b)

$\frac{dy}{dx} = 4\ln 5 \times 5^x(5^x - 1)^3$

18 c)

$\frac{dy}{dx} = \frac{4}{x}(\ln x)^3$

18 d)

$\frac{dy}{dx} = 6xe^{3x^2-1}$

18 e)

$\frac{dy}{dx} = (2x+5)\ln 4 \times 4^{x^2+5x+1}$

18 f)

$\frac{dy}{dx} = -\ln 5 \times 4x \times 5^{2x^2+1}$

19 a)

$\frac{dy}{dx} = 6 - 2e^{2x}$ stat pt when $\frac{dy}{dx} = 0$.

$$6 - 2e^{2x} = 0 \Rightarrow 6 = 2e^{2x} \Rightarrow e^{2x} = 3 \Rightarrow 2x = \ln 3 \Rightarrow x = \tfrac{1}{2}\ln 3$$

$$y = 6(\tfrac{1}{2}\ln 3) - e^{2\times\frac{1}{2}\ln 3} = 3\ln 3 - 3$$

Stat pt is $(\frac{1}{2}\ln 3, 3\ln 3 - 3)$.

19 b)

$\frac{dy}{dx} = \frac{2(1-2x)}{x-x^2}$ stat pt when $\frac{dy}{dx} = 0$.

$$\frac{2(1-2x)}{x-x^2} = 0 \Rightarrow 2(1-2x) = 0 \Rightarrow x = \tfrac{1}{2}$$

$$y = 2\ln(\tfrac{1}{4}) = -2\ln 4$$

Stat pt is $(\frac{1}{2}, -2\ln 4)$.

20 a)

$\frac{dy}{dx} = \frac{6x^2+2x}{2x^3+x^2-1}$

20 b)

$\frac{dy}{dx} = \frac{\frac{1}{2}x^{-1/2}}{x^{1/2}} = \frac{1}{2}x^{-1}$

20 c)

$\frac{dy}{dx} = \frac{-\frac{1}{2}x^{-3/2}}{x^{-1/2}} = -\frac{1}{2}x^{-1}$

20 d)

$\frac{dy}{dx} = \frac{\cos x}{\sin x} = \cot x$

20 e)

$\frac{dy}{dx} = \frac{e^x}{e^x+1}$

20 f)

$\frac{dy}{dx} = \frac{1}{x\ln x}$

21 a)

$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}\sec^2\sqrt{x}$

21 b)

$\frac{dy}{dx} = (\cos x - 2x)\sec^2(\sin x - x^2)$

21 c)

$\frac{dy}{dx} = \sec^2 x \; e^{\tan x}$

21 d)

$\frac{dy}{dx} = (\frac{1}{x} + 2e^{2x}) \sec^2(\ln x + e^{2x})$

21 e)

$\frac{dy}{dx} = \sec^2 x \; (\sec^2(\tan x))$

22)

$$\frac{dy}{dx} = \frac{\sec^2 x}{\tan x} = \frac{1}{\cos^2 x} \times \frac{\cos x}{\sin x} = \frac{1}{\cos x} \times \frac{1}{\sin x} = \sec x \operatorname{cosec} x \text{ as req.}$$

23)

$\frac{dy}{dx} = 2x \sec^2(x^2)$. Since $\sec^2(x^2) \geq 1$ and $x > 0$ it is always increasing.

24)

$y = (\tan(ax))^2$ so $\frac{dy}{dx} = 2a \sec^2(ax) \tan(ax)$.

$$2a \sec^2(1.5a) \tan(1.5a) = 0$$

So $a = 0$ but $a > 0$, or $\sec^2(1.5a) = 0$ no solutions, or $\tan(1.5a) = 0 \Rightarrow 1.5a = \pi, 2\pi, 3\pi$.

$$a = \tfrac{2}{3}\pi \text{ or } a = \tfrac{4}{3}\pi \text{ or } a = 2\pi$$

So $a \in \{\frac{2}{3}\pi, \frac{4}{3}\pi, 2\pi\}$.

25 a)

Surface area: $30\pi = \pi r h$ so $h = \frac{30}{r}$. Relationship between lengths: $r^2 + y^2 = h^2$.

$$r^2 + y^2 = (\tfrac{30}{r})^2 \text{ so } y = \sqrt{\tfrac{900}{r^2} - r^2} = \tfrac{1}{r}\sqrt{900 - r^4}$$

$$\begin{aligned} V &= \tfrac{1}{3}\pi r^2 \times \tfrac{1}{r}\sqrt{900 - r^4} = \tfrac{1}{3}\pi r \sqrt{900 - r^4} \\ &= \tfrac{1}{3}\pi \sqrt{900r^2 - r^6} = \tfrac{1}{3}\pi (900r^2 - r^6)^{1/2} \end{aligned}$$

25 b)

$$\frac{dV}{dr} = \tfrac{1}{6}\pi (1800r - 6r^5)(900r^2 - r^6)^{-1/2}$$

Stat pt when $\frac{dV}{dr} = 0$.

$$\tfrac{1}{6}\pi (1800r - 6r^5)(900r^2 - r^6)^{-1/2} = 0$$

$$\frac{\pi r(1800 - 6r^4)}{6\sqrt{900r^2 - r^6}} = 0 \Rightarrow 1800 - 6r^4 = 0 \Rightarrow 1800 = 6r^4 \Rightarrow r^4 = 300 \Rightarrow r = \sqrt[4]{300}$$

$$V = \tfrac{1}{3}\pi (900(\sqrt[4]{300})^2 - (\sqrt[4]{300})^6)^{1/2} = 106.75409835\ldots$$

$v = 107 \text{ cm}^3$ (3 SF)